

AI for Bharat Hackathon

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Team Name : MEGAVERS

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Problem Statement : AI-Powered Multilingual Public Grievance Classification & Resolution Support System for India

Brief about the Idea: The *AI-Powered Multilingual Public Grievance Classification & Resolution Support System for India* is a smart digital platform designed to streamline the grievance redressal process across diverse linguistic regions of the country. The system enables citizens to submit grievances in multiple Indian languages through text or voice. Using Artificial Intelligence and Natural Language Processing (NLP), the platform automatically detects the language, translates the grievance, classifies it into the appropriate category, predicts urgency, and routes it to the relevant government department. This ensures faster resolution, improved transparency, and inclusive access to public services.

Key Features:

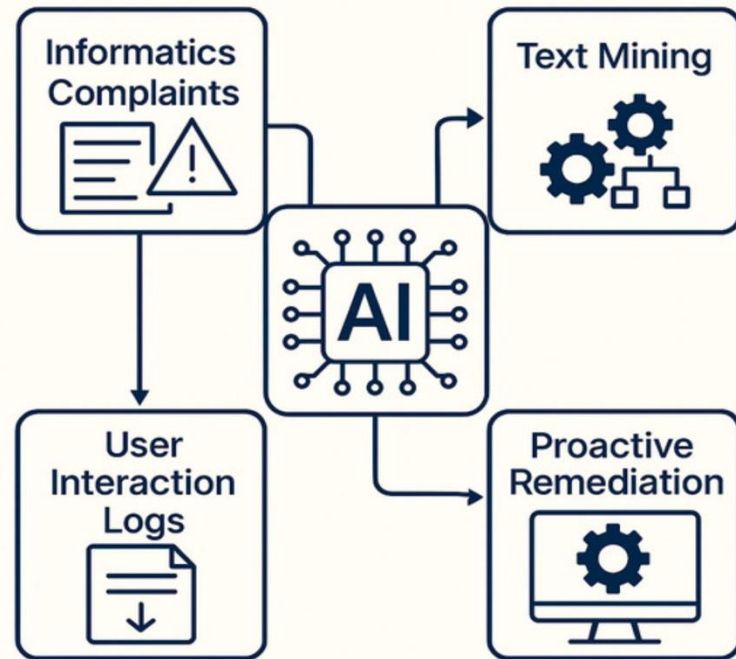
- Supports **multilingual grievance submission** across major Indian languages
- **Automatic language detection and translation** using NLP models
- **AI-based grievance classification** into sectors like health, education, sanitation, water, electricity, and transport
- **Priority and urgency prediction** to handle critical issues efficiently
- **Department-wise routing** for accurate and faster resolution
- **Real-time status tracking and notifications** for citizens
- **Admin dashboard and analytics** for government officials

The system leverages fine-tuned transformer-based models such as IndicBERT for multilingual text classification, Whisper for speech-to-text conversion, and IndicTrans2 for neural machine translation. The models are trained on public grievance datasets and Indian language corpora to ensure domain-specific accuracy.

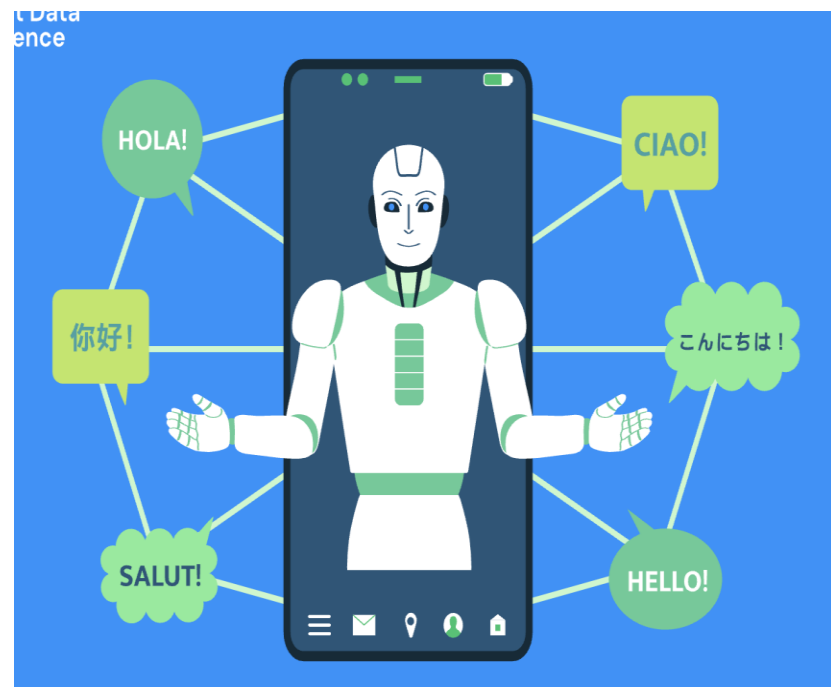
1. Unlike traditional grievance systems that rely on manual processing and limited language support, our solution uses **end-to-end AI automation** to handle grievances submitted in **multiple Indian languages (text and voice)**, with intelligent classification, prioritization, and department routing.
2. The system removes language and processing delays by **automatically detecting, translating, categorizing, and prioritizing grievances**, ensuring they reach the **right department faster**, while providing **real-time tracking and transparency** to citizens.
3. A **single unified AI platform** that combines **multilingual accessibility, urgency prediction, and smart routing** on a scalable cloud architecture, enabling **inclusive, efficient, and transparent grievance redressal** at a national scale.

Technical Differentiator:

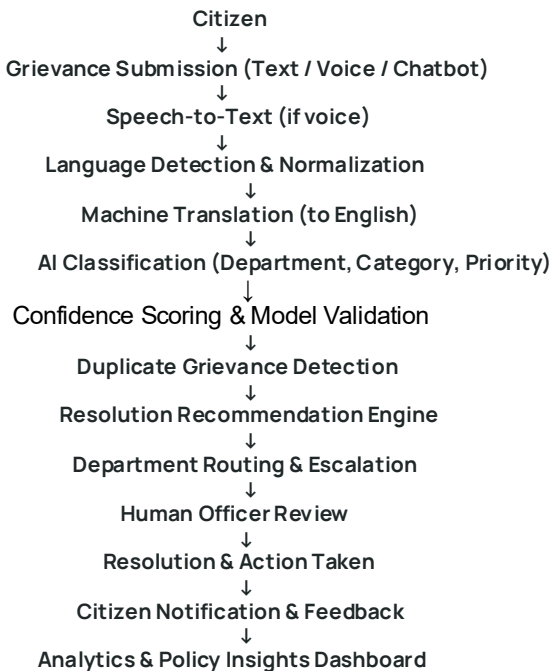
- Dialect-aware classification using language embeddings
- Smart escalation prediction based on historical resolution patterns
- AI-generated draft responses to reduce officer workload



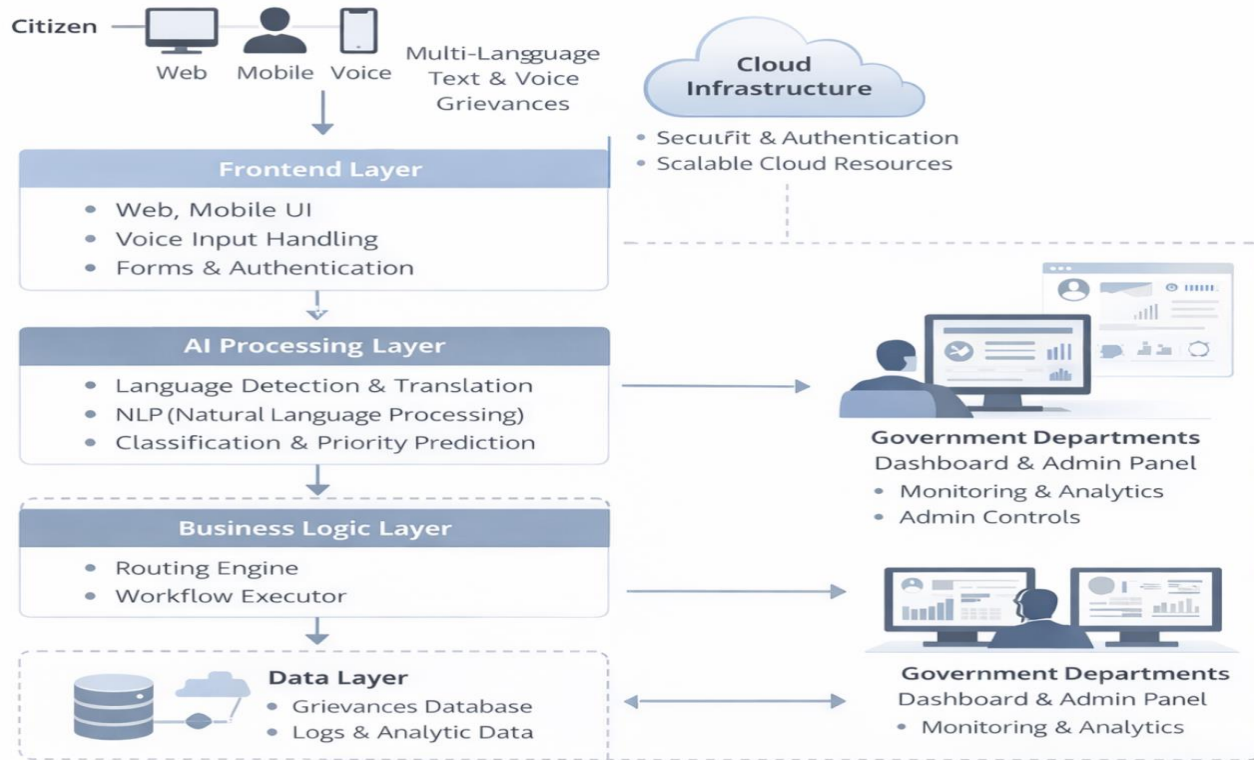
- **List of features offered by the solution**
 - Multilingual grievance submission (supports major Indian languages)
 - Voice and text-based input for inclusive access
 - Automatic language detection and translation using NLP
 - AI-based grievance classification into relevant sectors
 - Priority and urgency prediction for critical complaints
 - Intelligent department routing to avoid misclassification
 - Real-time grievance status tracking & notifications
 - Admin dashboard with analytics and insights
 - Scalable and secure cloud-based architecture
 - Duplicate grievance detection using Sentence-BERT embeddings
 - AI-assisted resolution summary generation
 - Semi-supervised model fine-tuning for low-resource languages



Process flow diagram



Architecture diagram of the proposed solution:



Technologies to be used in the solution:

Frontend Technologies

- HTML, CSS, JavaScript
- React / Angular for responsive web interface
- Mobile-friendly UI for citizen access

Backend Technologies

- Python (Flask / FastAPI) for API development
- RESTful APIs for system communication
- Authentication & Authorization mechanisms

AI / Machine Learning

- IndicTrans2 (Neural Machine Translation)
- Fine-tuned IndicBERT (Multilingual Classification)
- Whisper ASR (Speech Recognition)
- Sentence-BERT (Duplicate Detection)
- FastText (Language Identification)

Database & Storage

- Relational Database (MySQL / PostgreSQL)
- NoSQL Database (MongoDB) for scalable data storage
- Cloud Storage for logs and documents

Cloud & Deployment

- AWS Cloud Services
 - Compute (EC2 / Lambda)
 - Storage (S3)
 - Database (RDS)
- Scalable & Secure Cloud Architecture

Monitoring & Analytics

- Admin Dashboard
- Logging & Monitoring Tools
- Analytics for grievance trends and performance

Add as per the requirements for the hackathon:

- The solution directly addresses a **public governance challenge in India** by improving the grievance redressal system using AI
- Designed with **Bharat-scale inclusivity**, supporting **multiple Indian languages** and voice-based grievance submission
- Uses **Artificial Intelligence and NLP as the core**, not as an add-on, fulfilling the hackathon's AI-first requirement
- Focuses on **social impact, accessibility, and citizen empowerment**, especially for linguistically diverse populations
- Built to be **scalable, secure, and cloud-ready**, suitable for nationwide deployment
- Promotes **transparency, efficiency, and faster resolution** in public service delivery
- Strongly aligns with the **AI for Bharat vision of using AI for public good and governance transformation**

Expected Performance Metrics:

- Classification Accuracy $\geq 88\%$
- F1 Score ≥ 0.85 across major languages
- Routing Precision $\geq 90\%$
- ASR Word Error Rate $\leq 18\%$
- 40% reduction in grievance routing time

Dataset & Evaluation Strategy

- Public grievance data from CPGRAMS & state portals
- AI4Bharat language corpora
- Target dataset size: 50,000–100,000 samples
- Semi-supervised labeling with manual validation
- Cross-language evaluation

Innovation partner **I12S**

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Thank You

